

Roman Numeral to Integer Converter

Two Java Classes

Question

Create a Java class that converts an uppercase Roman numeral to its integer value, then write a second class that reads a numeral from the user and displays the conversion result.

How It Works

1. The `Roman` class maps each Roman symbol to its integer value.
2. It scans the input from left to right. If a symbol is smaller than the symbol immediately after it, subtract its value; otherwise, add its value.
3. The `ImportRoman` class reads the text from the console, calls `romanToInteger()`, and prints the result.

Files and Package

Save the files using this layout. The package declaration means `Roman.java` must be inside the `number` directory.

```
project-folder/  
|-- ImportRoman.java  
|-- number/  
    |-- Roman.java
```

Source: `number/Roman.java`

```
package number;  
  
public class Roman {  
  
    public int romanToInteger(String s) {  
        int total = 0;  
  
        for (int i = 0; i < s.length(); i++) {  
            int current = value(s.charAt(i));  
  
            if (i + 1 < s.length() &&  
                current < value(s.charAt(i + 1))) {  
                total = total - current;  
            } else {  
                total = total + current;  
            }  
        }  
  
        return total;  
    }  
  
    // Returns value of Roman symbol  
    private int value(char ch) {  
        switch (ch) {  
            case 'I': return 1;  

```

```
        case 'V': return 5;
        case 'X': return 10;
        case 'L': return 50;
        case 'C': return 100;
        case 'D': return 500;
        case 'M': return 1000;
        default: return 0;
    }
}
```

Source: ImportRoman.java

```
import number.Roman;
import java.util.Scanner;

public class ImportRoman {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roman Number: ");
        String roman = sc.nextLine();

        Roman r = new Roman();

        int result = r.romanToInteger(roman);

        System.out.println("Integer Value: " + result);

        sc.close();
    }
}
```

Compilation and Execution

From the project folder, compile both files and run the main class:

```
javac number/Roman.java ImportRoman.java
java ImportRoman
```

Sample Run

```
Enter Roman Number: MCMXCIV
Integer Value: 1994
```

The converter expects uppercase Roman symbols (I, V, X, L, C, D, and M) and assumes the input is a valid Roman numeral; it does not validate numeral formatting.